

Literature-Already-Answered Packet

The pair (c_0, c_0) does not have the CPPm

Status: literature already answered, not a new automatic-discovery result

Source and later answer

The source paper is:

arXiv:2601.17316, Manwook Han, *Weak minimizing property on pairs of classical Banach spaces*.

The local copy is included as `source_paper.pdf`.

The later supporting paper is:

arXiv:2605.01397, Anselmo Raposo Jr. and Geivison Ribeiro,

Weak Minimizing Property and the Compact Perturbation Property for the Minimum Modulus.

The local copy is included as `supporting_paper_2605.01397.pdf`.

Open question location

The following full-width crop is taken from page 13 of the source PDF. It shows Han's statement that the case of (c_0, c_0) remained undecided and the exact Question 3.6.

has property (M) . In particular, it satisfies the WMP. However, by Example 3.5, this pair fails the CPPm. Consequently, it fails both property (m) and the WmP.

We have now obtained nearly complete information on which pairs of separable classical sequence spaces satisfy the WmP or the CPPm, and which do not. Nevertheless, despite our efforts, we were unable to decide whether the pair (c_0, c_0) has the CPPm, and we conclude the paper by posing this as an open problem.

Question 3.6. Does the pair (c_0, c_0) have the CPPm?

ACKNOWLEDGEMENTS

The author would like to thank Sun Kwang Kim for helpful discussions and valuable comments.

Question

Han asks:

Does the pair (c_0, c_0) have the CPPm?

Here CPPm is the compact perturbation property for the minimum modulus: for every operator $T : X \rightarrow Y$ that does not attain its minimum modulus,

$$\sup_{K \in \mathcal{K}(X, Y)} m(T + K) = m(T), \quad m(T) = \inf \{\|Tx\| : x \in S_X\}.$$

Answer

Raposo–Ribeiro explicitly cite Han’s Question 3.6 and answer it negatively. The following is the constructive proof, rewritten in a self-contained form.

Theorem. *The pair (c_0, c_0) does not have the CPPm.*

Proof. Let

$$f(x) = \sum_{j=2}^{\infty} 2^{1-j} x_j, \quad x = (x_j)_{j \geq 1} \in c_0.$$

Then $f \in (c_0)^* = \ell_1$, $\|f\| = 1$, $f(e_1) = 0$, and f has infinite support. Define

$$T : c_0 \rightarrow c_0, \quad Tx = x - f(x)e_1.$$

We first compute the minimum modulus of T . Let $x \in S_{c_0}$ and put

$$s = \sup_{j \geq 2} |x_j|.$$

Since f only sees the coordinates $j \geq 2$, we have $|f(x)| \leq s$, and

$$\|Tx\| = \max\{|x_1 - f(x)|, s\}.$$

If $s \geq 1/2$, then $\|Tx\| \geq 1/2$. If $s < 1/2$, then the equality $\|x\|_{\infty} = 1$ forces $|x_1| = 1$, and therefore

$$|x_1 - f(x)| \geq |x_1| - |f(x)| \geq 1 - s > 1/2.$$

Thus $\|Tx\| \geq 1/2$ for every $x \in S_{c_0}$.

For the reverse inequality, take

$$x^{(N)} = e_1 + \frac{1}{2} \sum_{j=2}^N e_j.$$

Then $\|x^{(N)}\|_{\infty} = 1$ and

$$f(x^{(N)}) = \frac{1}{2} \sum_{j=2}^N 2^{1-j} \rightarrow \frac{1}{2}.$$

Consequently

$$\|Tx^{(N)}\| = \max \left\{ \left| 1 - f(x^{(N)}) \right|, \frac{1}{2} \right\} \rightarrow \frac{1}{2},$$

so $m(T) = 1/2$.

Next we show that T does not attain its minimum modulus. Suppose, toward a contradiction, that $\|x\|_{\infty} = 1$ and $\|Tx\| = 1/2$. Then $s \leq 1/2$. Since $\|x\|_{\infty} = 1$, this implies $|x_1| = 1$. Multiplying x by a scalar of modulus one if necessary, we may assume $x_1 = 1$.

The inequality $\|Tx\| = 1/2$ gives

$$|1 - f(x)| \leq \frac{1}{2},$$

hence $|f(x)| \geq 1/2$. On the other hand,

$$|f(x)| \leq \sum_{j=2}^{\infty} 2^{1-j} |x_j| \leq \frac{1}{2} \sum_{j=2}^{\infty} 2^{1-j} = \frac{1}{2}.$$

Therefore equality holds throughout. Since all weights 2^{1-j} are positive, equality in the second inequality forces $|x_j| = 1/2$ for every $j \geq 2$. This contradicts $x \in c_0$. Hence T is not minimum-modulus attaining.

Finally define the rank-one operator

$$K : c_0 \rightarrow c_0, \quad Kx = f(x)e_1.$$

It is compact, and

$$(T + K)x = x \quad (x \in c_0).$$

Thus $T + K = I_{c_0}$, so

$$m(T + K) = 1 > \frac{1}{2} = m(T).$$

This violates the defining CPPm equality for the non-minimum-modulus-attaining operator T . Therefore (c_0, c_0) does not have the CPPm. $\square$

Classification

This packet belongs in `solutions/literature_already_answered/`. The later paper explicitly states that Han left Question 3.6 open and that it gives a negative answer. Therefore this is a successful retrieval and verification hit, not an original automatic-discovery proof.

Verification code

The script `code/verify_cppm_example.py` checks the finite truncations of the minimizing sequence $x^{(N)}$ and verifies the perturbation identity $T + K = I$ on sample finite vectors. The computation is not a substitute for the analytic non-attainment argument; it is only a reproducibility sanity check for the construction.

Reproducibility

The screenshot crop was generated from the included source PDF by

`code/crop_open_question.py`.

All LaTeX build products are placed in `tmp/`. The final rendered packet is `solution_packet.pdf`.